

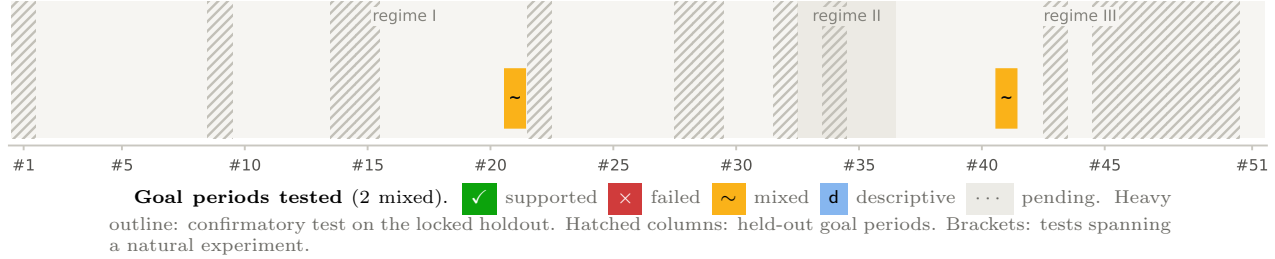
H24: In the forecast week (#21), switching from independent drafts to comparison switches the coupling on

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 1.5/5

Bottom line. In the forecast week the coupling switch was real but early and moved no swarm-level alignment beyond ordinary kickoff days, in either embedding model; the coupling that exists is pairwise and gated by reading, about 0.03 in cosine per document read.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Goal period #21 (forecast AI; regime I; 8–9 agents) as a natural experiment. From the record, each agent first read a teammate’s forecast 42–86 min into the week (median 50 min), not mid-week; the share of messages naming a teammate jumps there from 7% to 74%. The predicted alignment step did not appear: $\Delta A_{\text{res}} = +0.010 [-0.05, +0.09]$, inside a within-week placebo and 23 other kickoff days. Content was aligned before any reading ($A_{\text{res}} \approx 0.33$ vs. a rotation null of 0.03). Numeric herding was an extraction artifact (hand-verified $\kappa = 0.25$, $p = 0.17$). **Round 1b** (second embedding model, shared and multi-direction goal field, dedupe, style): the step fails in all 13 configurations, but gte sees +0.16 that ordinary kickoff days match. Natives: reading a teammate’s document pulls the reader toward it by ≈ 0.03 ; one kickoff split into two rooms grew two room cultures. Holdout (#14) not run.

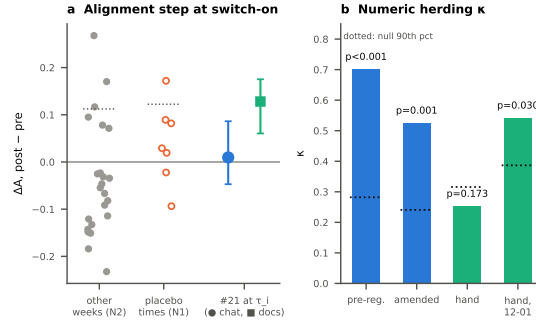
PHYSICAL MODEL

Model 11 (vector spins), with model 01’s Curie–Weiss analogue. Spin $\mathbf{s}_i$: agent i ’s whitened, unit-normalized statement embedding ($n = 32$). Field $h\hat{g}$: the goal and kickoff text. $\mathbf{h}_i$: agent style. Coupling J_0 : a mean-field pull toward the others, switched on at τ_i (the agent’s first reading of a teammate’s forecast). A : mean pairwise cosine, with $\hat{g}$ removed for A_{res} . Numeric forecasts x_i (AGI by 2035, SI by 2050, p(doom)) are soft spins with a DeGroot pull κ .

$$\mathbf{s}_i(t) \propto h\hat{g} + \mathbf{h}_i + \xi_i(t) + J_0 \Theta(t - \tau_i) \mathbf{m}_{-i}(t)$$

$$A = \frac{2}{N(N-1)} \sum_{i < j} \hat{\mathbf{v}}_i \cdot \hat{\mathbf{v}}_j, \quad \beta J_0/n = 1 - \frac{1}{1 + (N-1)A_{\text{res}}}$$

$$\Delta x_i = \kappa (\bar{x}_{-i} - x_i) + \epsilon_i, \quad \kappa = 0 \text{ for independent updating}$$



Observables. (a) Change in residual alignment across the switch-on for #21 (blue: statements; green: documents; 90% CIs), against 23 other regime-I kickoff days at the same offset (gray) and placebo switch times inside #21 (orange). Dotted lines: 90th percentiles. (b) DeGroot pull κ under the pre-registered extractor, the amended extractor and hand-verified values; dotted: the independent-updating null, 90th percentile. The herding signal shrinks to noise once extraction errors are removed.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly a cautionary tale. Telling LLM agents to “form independent views, then compare” did not buy independence: their content was aligned before anyone read anyone, from the shared prompt, instructions and prior. Comparison then produced talk about divergence, and some literal copying of numbers, not a measurable alignment jump. For anyone aggregating swarm forecasts, an independent-first-draft phase is no diversity guarantee: most of the diversity worth protecting is gone at kickoff. The reusable lessons are methodological. Regex-extracted numbers manufacture herding through regression to the mean, and snapshot coupling estimates absorb unmodelled fields. One week, $N = 7$, no ground truth: a pilot.

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RESULTS VS PREDICTIONS

Prediction	Round 1 → 1b (bge / gte)	Verdict
P1 step > q90 of N1 and N2	+0.010 → -0.011 / +0.160 (N2 q90 × 0.10 / 0.18); 13/13 fail	×
P2 documents $\Delta A_{\text{res}} > 0$	+0.13 → +0.16 / +0.08 (CIs > 0)	✓
P3 $\beta J_0/n$ rises	0.67→0.67 → flat / 0.69→0.76	model-dep.
P4 numeric herding κ 0.25, p 0.17 (no embedding) (audited)		×
P6 ramp $\rho > 0$	0.73 → 0.47 / 0.60	~
Levels vs rotation null	0.33 vs 0.03 → 0.34–0.42 vs ≤0.10	✓
G21 reading DiD (na- statements +0.036 (p 0.035) / ✓ tive)	+0.030 (p 0.038); style p 0.08–0.14	✓
G41 two rooms (na- no gap in first drafts; gap grows to ~ tive)	+0.25 / +0.32 by day 4	~

G21 ~ 1b: mixed (step fails; pair DiD supported)

G41 ~ mixed (round 1b native; both models)

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	0	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

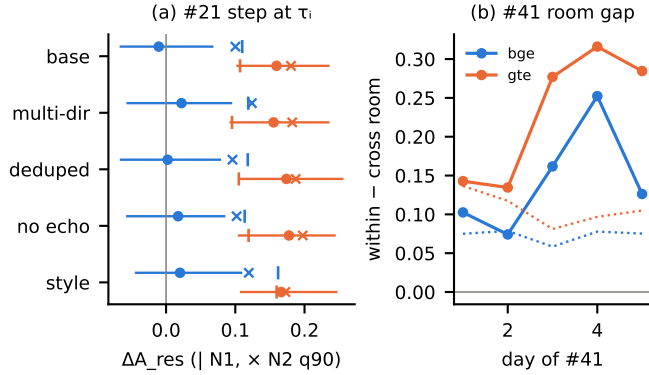
One forecasting week, $N = 7$; the switch-on and the reading events rest on agents' own screen descriptions. The step statistic is embedding-model dependent. The pair DiD weakens after style residualization, so part of it may be register. In #41 everyone reads room-mates within a minute, so early coupling and room composition are not separable; only 9 one-statement drafts.

ROUND 2

Copy fidelity of numbers after reading (H07 machinery) and an anchoring test (first-posted numbers dominate), per the round-2 redirects.

- **Essence.** Information aggregation in swarms is copying from the first or most confident agent, not averaging.
- **H24-R1** Anchoring cascade: first-posted numbers dominate the final consensus (a founder effect) in every estimation-like task.
- **H24-R2** Copy fidelity vs transformation of shared numbers (H07 machinery).

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Second figure. (a) Step at the switch-on per input (blue bge, orange gte; bars 90% CI; | within-week placebo q90, × kickoff-matched q90): gte steps up, but no more than ordinary kickoff days. (b) #41: within-room minus cross-room alignment by day (dotted: permutation q95).